

Project Progress Report

Generate Anime Faces with GANs:
A Deep Learning Approach to Stylized Face
Synthesis

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Link to Github: github.com/ZudzinHuang/anime-face-generation-gan

Project Description

This project focuses on generating 64×64 RGB anime faces using Generative Adversarial Networks (GANs). The input is a latent noise vector, and the output is a synthetic anime-style face image. The goal is to train a model that can learn visual patterns from anime face images, such as hair, eyes, face shape, and colour style, then generate new faces that look visually consistent with the dataset. This project is interesting because it shows whether a model can learn complex visual structure without being directly told rules for how an anime face should look. Deep learning is appropriate for this task because the patterns in images are too complex to define manually, while GANs are designed to learn image distributions and generate realistic synthetic images. So far, the project uses a DCGAN baseline and a WGAN-GP primary model, with generated samples and loss curves used to compare progress [1, 2].

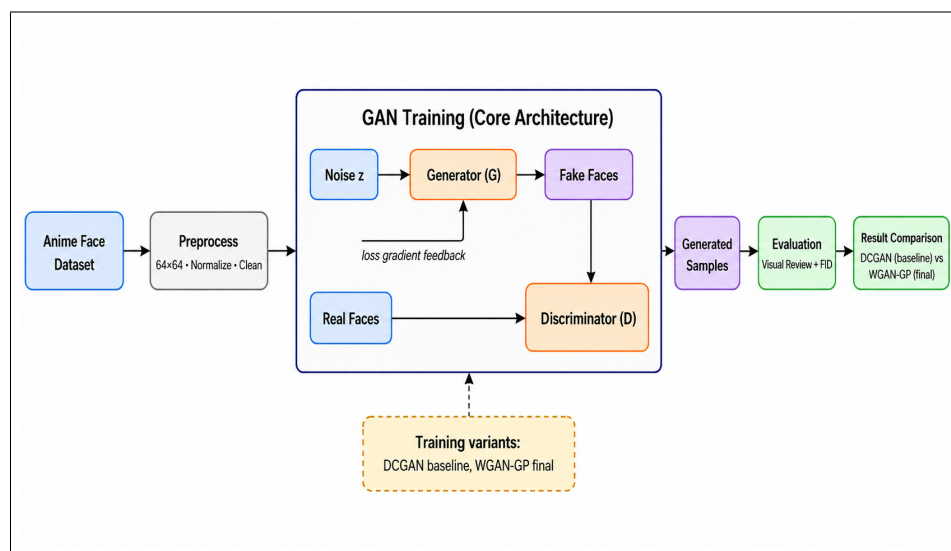
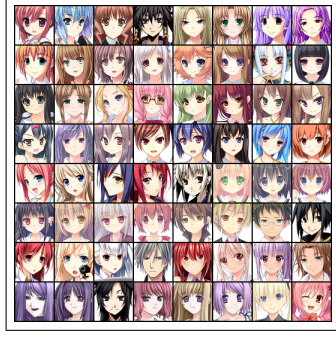


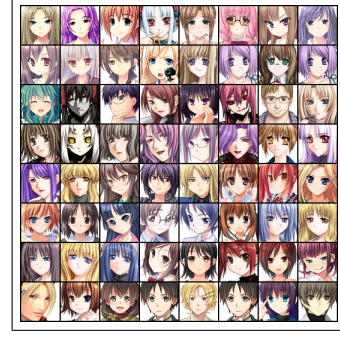
Figure 1: Project workflow: anime face dataset preprocessing, DCGAN baseline training, WGAN-GP primary model training, sample generation, and evaluation using visual review and possible quantitative metrics such as FID.

Data Processing

The dataset is the Kaggle Anime Faces dataset [3]. All images were converted to RGB, centre-cropped, resized to 64×64 , saved as PNGs, then normalized with mean/std (0.5, 0.5, 0.5) so pixel values match the generator's $\tanh$ output range. After cleaning, the dataset contained 21,551 usable images. Using seed 360, it was split into 17,241 training images and 4,310 held-out test images. Models are trained only on the training split, while the held-out split provides never-before-seen real images for final comparison. Since GANs do not use labels or accuracy, progress is judged using real samples, generated samples, and loss curves. Figures 2a and 2b show training and held-out test samples.



(a) Cleaned real training samples.

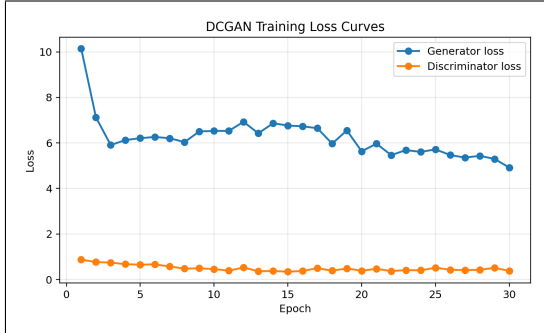


(b) Held-out real test samples from the 20% test split.

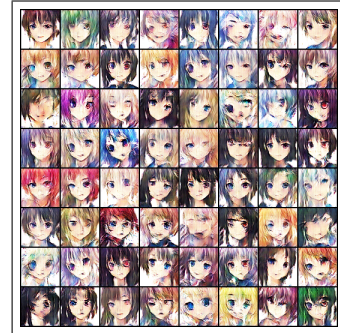
Figure 2: Examples from the processed anime face dataset.

Baseline Model: DCGAN

The baseline is a DCGAN, a CNN-based GAN for small image generation [1]. The generator maps a 100-dimensional latent vector into a 64×64 RGB image, while the discriminator classifies real and generated images. This is a reasonable baseline because it is a standard architecture for comparing against the stronger WGAN-GP objective. DCGAN was trained for 30 epochs with batch size 128, Adam optimizer, learning rate 0.0002, and betas (0.5, 0.999). It has 3,576,704 generator parameters and 2,765,568 discriminator parameters. At epoch 30, generator loss was 4.919 and discriminator loss was 0.382. Qualitatively, DCGAN learned rough anime-face structure, including face colour, hair regions, and eye placement, but many samples still have distorted features. Figures 3a and 3b show the loss curves and final generated samples.



(a) DCGAN loss curves.



(b) DCGAN final generated samples using fixed seed 360.

Figure 3: DCGAN baseline quantitative and qualitative results.

Primary Model: WGAN-GP

The primary model is WGAN-GP, a convolutional GAN that replaces DCGAN’s binary cross-entropy objective with a Wasserstein-based objective and gradient penalty [2]. This fits the task because the gradient penalty helps stabilize GAN training and reduce mode collapse, which is important for generating consistent anime faces. The model takes a 100-dimensional latent vector and generates a 64×64 RGB anime face. Both networks use convolutional blocks: the generator upsamples the latent vector through transposed-convolution layers with batch normalization and ReLU before a final tanh output layer, while the critic downsamples images through convolutional layers with LeakyReLU before producing one real-valued score. As shown in Table 1, it was trained for 30 epochs using Adam, learning rate 0.0001, gradient penalty $\lambda = 10$, and 5 critic updates per generator update. At epoch 30, WGAN-GP reached a generator loss of 1.990, critic loss of -11.123, and gradient penalty of 0.173. Qualitatively, Figure 4 shows clearer colours and more coherent face structure than the DCGAN baseline, although artifacts such as warped eyes, unclear mouths, and repeated front-facing poses remain. Since DCGAN and WGAN-GP use different loss objectives, the models are compared mainly through training stability and generated sample quality.

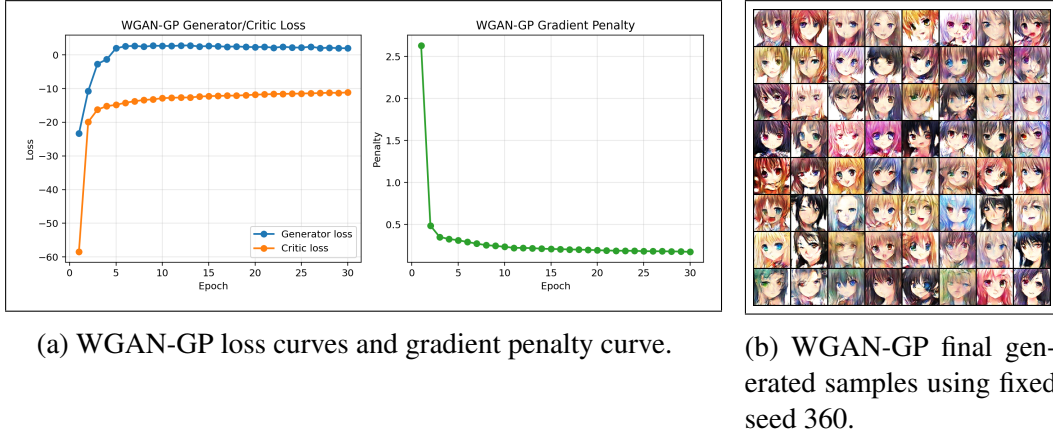


Figure 4: WGAN-GP primary model quantitative and qualitative results.

Table 1: Model summary comparing DCGAN and WGAN-GP.

Category	DCGAN Baseline	WGAN-GP Primary Model
Input	100D latent vector	100D latent vector
Output	64×64 RGB image	64×64 RGB image
Training epochs	30	30
Batch size	128	64
Optimizer	Adam	Adam
Learning rate	0.0002	0.0001
Betas	(0.5, 0.999)	(0.0, 0.9)
Generator parameters	3,576,704	3,576,704
Discriminator/Critic parameters	2,765,568	2,763,776
Final generator loss	4.919	1.990
Final discriminator/critic loss	0.382	-11.123
Extra metric	N/A	Gradient penalty: 0.173

References

- [1] A. Radford, L. Metz, and S. Chintala, “Unsupervised representation learning with deep convolutional generative adversarial networks,” *arXiv preprint arXiv:1511.06434*, 2016.
- [2] I. Gulrajani, F. Ahmed, M. Arjovsky, V. Dumoulin, and A. C. Courville, “Improved training of wasserstein gans,” *arXiv preprint arXiv:1704.00028*, 2017.
- [3] S. Rakshit, “Anime faces dataset.” <https://www.kaggle.com/datasets/soumikrakshit/anime-faces>, n.d. Kaggle.